



EXPERIMENT 4

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Subject Name: Computer Network

Aim : To create a multi-router network topology in Cisco Packet Tracer and implement static routing for communication between separate LANs. The experiment also involves testing network connectivity to ensure successful data transfer across all connected devices.

Objective

- To gain practical knowledge of static routing and understand its role in enabling communication between different networks.
- To assign and configure appropriate IP addresses on PCs, router interfaces, and switches (where necessary) using Cisco Packet Tracer.
- To establish connectivity between multiple networks by manually creating and configuring static routes on each router.
- To test and validate the network configuration by performing end-to-end connectivity checks using tools such as the ping command.

Theory:

Static routing is a routing method in which the network administrator manually configures the paths that data packets should follow to reach their destination. In this approach, routing information is entered directly into the routing table of each router using commands that specify the destination network and the corresponding next-hop IP address or exit interface. Since the routes are manually configured, routers do not exchange routing information with one another or automatically update their routing tables when changes occur in the network topology.

Unlike dynamic routing protocols such as RIP, OSPF, or EIGRP, static routing does not adapt to network failures or topology changes. If a link fails or a new network is added, the administrator must manually modify the routing configuration to restore connectivity. Because of this, static routing is best suited for small, stable networks where the network structure changes infrequently.

One of the major advantages of static routing is its simplicity. It consumes very little CPU processing power, memory, and network bandwidth because routers do not need to exchange routing updates. It also provides better security, as routes can only be modified by an authorized administrator, reducing the risk of unauthorized routing changes. Furthermore, static routing offers predictable routing paths, making network behavior easier to understand and troubleshoot.

However, static routing also has certain limitations. As the network grows in size and complexity, manually configuring and maintaining routes on every router becomes time-consuming and prone to human error. In large enterprise networks, this lack of scalability makes static routing less practical compared to dynamic routing protocols, which automatically learn and update routes.

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Practical Steps:

Step 1: Create LAN 1

- Place 1 Switch, 2 PCs, and 1 Router (Router3) in the Cisco Packet Tracer workspace.
- Connect both PCs to the switch using Copper Straight-Through cables.
- Connect the switch to Router3 through the GigabitEthernet0/0 interface using a Copper Straight-Through cable.
- Ensure that all devices are properly connected before proceeding to the next step.

Step 2: Create LAN 2

- Place 1 Switch, 2 PCs, and 1 Router (Router4) in the workspace.
- Connect both PCs to the switch using Copper Straight-Through cables.
- Connect the switch to Router4 via the GigabitEthernet0/0 interface.
- Verify that all cable connections are correctly established.

Step 3: Create LAN 3

- Place 1 Switch, 2 PCs, and 1 Router (Router5) in the workspace.
- Connect both PCs to the switch using Copper Straight-Through cables.
- Connect the switch to Router5 using the GigabitEthernet0/0 interface.
- Check that each device is connected properly to avoid connectivity issues during configuration.

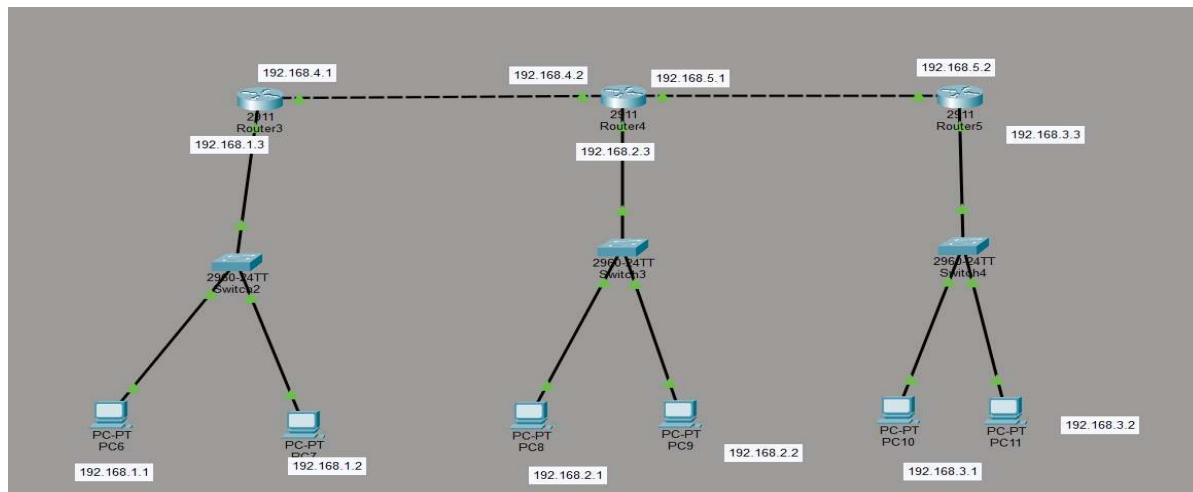
Step 4: Connect the Routers

- Connect Router3 to Router4 using a Serial DCE/DTE cable.
- Connect Router4 to Router5 using another Serial DCE/DTE cable.
- Ensure that one end of each serial connection is configured as the DCE end so that a clock rate can be assigned if required.

Step 5: Verify Router Interconnections

- Recheck the serial connections between Router3 and Router4 and between Router4 and Router5.
- Confirm that the correct serial interfaces have been used and that the cables are securely connected.
- Once all physical connections are complete, the network topology is ready for IP addressing and static routing configuration.

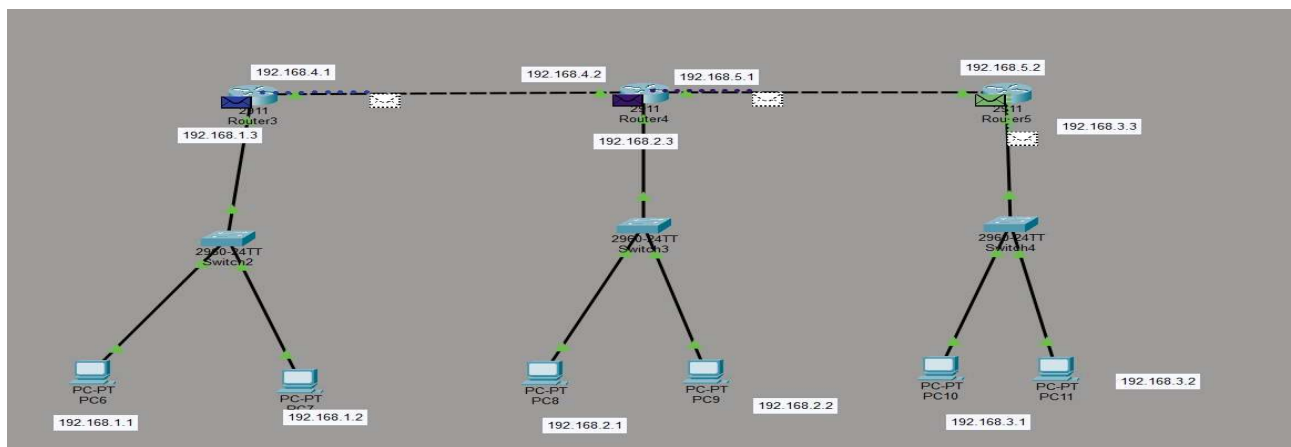
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Step 6: Configure Static Routing

Configure static routes on each router so that all three LANs can communicate.

- Give Network Address of Network 192.168.2.0, 192.168.3.0 and their subnet mask and net hop to Router 3
- Give Network Address of Network 192.168.1.0, 192.168.3.0 and their subnet mask and next hop to Router 4
- Give Network Address of Network 192.168.1.0, 192.168.2.0 and their subnet mask and next hop to Router 5
- Connect Router4 to Router5 using another Serial DCE/DTE cable.



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Learning Outcome

After successfully completing this experiment, the learner will be able to:

1. Understand the fundamentals of static routing and explain how manually configured routing tables enable communication between multiple networks without relying on dynamic routing protocols. The learner will also be able to identify the advantages, limitations, and practical applications of static routing in small and stable network environments.
2. Develop practical skills in configuring network devices by assigning appropriate IP addresses, subnet masks, and default gateways to end devices, as well as configuring router interfaces using Cisco Packet Tracer. The learner will gain confidence in establishing proper addressing schemes required for successful network communication.
3. Acquire hands-on experience in implementing static routing by configuring static routes on multiple interconnected routers. The learner will understand how destination networks, subnet masks, and next-hop IP addresses work together to establish communication between separate LANs and ensure successful end-to-end data transmission.
4. Learn to verify, analyze, and troubleshoot network connectivity using various network testing and diagnostic tools such as the ping command, Simple PDU, show ip route, show running-config, and show ip interface brief commands. These tools will help the learner identify configuration errors, validate routing tables, and confirm successful communication across interconnected networks.
5. Improve problem-solving and network troubleshooting skills by identifying and correcting common configuration errors such as incorrect IP addressing, subnet mask mismatches, missing static routes, interface shutdown issues, and incorrect next-hop addresses. This practical experience strengthens the learner's ability to deploy and maintain small-scale routed networks.
6. Understand the importance of proper network planning and routing configuration in designing reliable and scalable computer networks. The learner will appreciate how accurate IP addressing and routing decisions contribute to efficient communication between geographically separated networks while preparing them for more advanced routing concepts such as RIP, OSPF, and EIGRP.